

WAKE ISLAND, United States Minor Outlying Islands

WMO: 912460

Lat: 19.283N

Lon: 166.650E

Elev: 7

StdP: 101.24

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 41606

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.4	22.7	14.1	10.1	23.7	15.6	11.0	24.4	14.4	23.3	12.9	24.1	6.9	50	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	32.7	26.5	32.3	26.5	31.7	26.4	28.6	29.8	28.0	29.2	27.6	28.9	5.0	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.1	28.8	27.7	23.7	28.4	27.4	23.2	28.2	91.8	30.1	90.2	28.4	87.4	29.6	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	11.1	10.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	25.9	25.7	25.9	26.8	27.7	28.8	28.9	29.0	29.0	28.5	27.6	26.7
	DBStd	1.57	1.02	1.10	0.86	0.96	1.01	0.87	0.99	1.04	0.92	0.97	0.86	0.90
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6405	493	438	492	504	548	563	584	590	570	573	529	518
	CDD18.3	3363	235	205	234	254	290	313	326	332	320	315	279	260
	CDH23.3	35021	1753	1427	1714	2336	3073	3711	3936	4088	3929	3723	2951	2380
CDH26.7	10455	215	172	225	489	862	1359	1507	1655	1564	1297	708	402	
Wind	WSAvg	5.8	5.7	6.1	6.7	6.5	5.4	5.2	5.5	4.9	5.0	6.0	7.1	6.0
Precipitation	PrecAvg	754	23	34	41	43	35	50	83	101	96	91	58	32
	PrecMax	1234	108	166	176	158	142	138	273	361	408	297	147	121
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	318	24	38	48	39	33	40	81	95	82	77	43	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.6	29.1	29.5	31.3	33.0	33.9	33.7	33.1	32.8	32.4	31.2	30.2
		MCWB	24.5	24.4	23.8	24.8	25.4	26.6	26.5	27.4	27.4	26.4	25.3	24.8
	2%	DB	29.0	28.8	28.9	30.2	31.3	32.4	32.4	32.4	32.3	31.5	30.3	29.5
		MCWB	23.8	24.1	23.5	24.6	25.5	26.2	26.4	26.9	26.9	26.3	25.2	24.7
	5%	DB	28.4	28.0	28.3	29.7	30.7	31.4	31.7	32.0	31.7	31.1	30.0	29.0
		MCWB	23.5	23.4	23.3	24.5	25.2	25.7	26.3	26.7	26.6	26.1	25.1	24.4
	10%	DB	27.8	27.7	27.8	29.0	30.0	31.0	31.2	31.3	31.2	30.6	29.3	28.6
		MCWB	23.2	23.2	23.1	24.2	25.1	25.5	26.2	26.5	26.5	25.9	24.9	24.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	25.7	25.6	26.2	27.4	28.0	28.5	30.7	29.7	28.6	28.0	26.3
		MCDB	27.6	27.8	26.9	27.9	28.4	29.3	30.0	31.4	30.3	29.6	28.2	27.6
	2%	WB	25.1	25.1	24.9	25.6	26.8	27.4	27.9	28.5	28.5	27.9	27.3	25.8
		MCDB	27.5	27.3	26.8	28.1	28.4	29.1	29.4	30.0	29.8	29.0	27.9	27.6
	5%	WB	24.6	24.6	24.4	25.2	26.2	26.9	27.5	27.8	27.9	27.5	26.5	25.4
		MCDB	27.2	27.0	26.8	28.1	28.4	29.2	29.0	29.2	29.2	28.7	28.0	27.6
	10%	WB	24.1	24.1	24.0	24.8	25.7	26.4	27.1	27.3	27.5	27.1	25.9	25.0
		MCDB	26.9	26.7	26.6	27.8	28.5	29.3	29.1	29.0	28.9	28.7	28.0	27.5
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.1	4.4	4.6	4.5	4.6	4.6	4.6	4.5	4.2	3.8	3.8
		MCDBR	4.4	4.4	4.7	5.3	5.4	5.3	5.2	5.2	5.0	4.8	4.2	4.3
		MCWBR	2.2	2.3	2.2	2.1	2.1	2.2	2.2	2.3	2.4	2.2	2.2	2.2
	5% WB	MCDBR	4.2	4.2	4.6	4.7	4.7	5.0	5.0	4.8	4.7	4.4	4.1	4.0
		MCWBR	2.3	2.3	2.5	2.3	2.4	2.6	2.6	2.9	2.8	2.6	2.5	2.2
Clear-Sky Solar Irradiance	taub	0.370	0.375	0.396	0.415	0.406	0.399	0.416	0.415	0.408	0.417	0.402	0.381	
	taud	2.582	2.558	2.506	2.474	2.523	2.556	2.516	2.518	2.545	2.495	2.504	2.547	
	Ebn at Noon	912	925	913	892	889	887	873	880	889	871	870	887	
	Edh at Noon	93	100	109	114	107	103	107	107	103	105	100	94	
All-Sky Solar Radiation	RadAvg	4.77	5.48	6.25	6.81	7.03	7.08	6.46	6.02	5.78	5.37	4.91	4.62	
	RadStd	0.19	0.30	0.19	0.33	0.23	0.32	0.49	0.61	0.43	0.28	0.20	0.15	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page